

Design, construction and test of an optimized Faraday cup for beam current determination of a helicon ion source

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ABSTRACT

Faraday cups are widely used to characterize ion beam current. A deep canonical Faraday cup for determining ion beam currents from a helicon ion source has been designed, built and tested with beam. In this paper, CST Studio code is used to design the Faraday cup. The precise operation of a Faraday cup relies on its capability to recapture the ejected electrons when energetic particles collide with interior cup. The interior geometry of the Faraday cup is designed to minimize the loss of particles. In addition, a biased ring is embedded in front of the cup collector at voltages up to -300 V with respect to the collector. The collector is made of copper while the suppressor ring and the Faraday cup body are made of stainless steel. The measurements are utilized at RF powers and extraction voltages up to 1300 W and 20 kV, respectively. The measured ion beam current is an indirect way to demonstrate mode transition from inductively coupling plasma (ICP) to helicon mode of operation in the constructed helicon ion source. We have detected a sharp increase in the measured ion beam as the mode transition is reached. In addition, Optical emission spectroscopy (OES) method was also employed for demonstrating the mode transition.

1. Introduction

A new helicon ion source has been recently developed at Amirkabir University of Technology for material processing applications [1]. Faraday cup is an instrument which can be used for beam current determination of different ion sources. The most important challenge in measuring the current through the Faraday cup is the loss of charged particles such as ion-induced secondary electrons and primary scattered ions from the inside of the cup. To overcome this problem several solutions have proposed by researchers as follows. In Ref. [2] an analysis of the losses caused by secondary particle emission was presented for different suppressor electrode voltages and beam energies. In Ref. [3and4] they consider the design of a deep ion collector cavity with large aspect ratio for reducing the loss of the particle. Applying electric and magnetic fields are the other ways that used in Refs. [5–7] to greatly reduce the escape of secondary electrons. In Ref. [8] they avoid loss of secondaries and backscatters by use of carbon to minimize the incidence cross-section. Embedment of a permanent magnet in the collector and improvement of collector geometry were the other methods which they employed. In Ref. [9and10] the suppressor grid with negative voltage and series of high-voltage grids as a particle filtering were used to protect the Faraday cup from the space plasma particles and return any secondary electrons to the collector. In Ref.

[11and12] an improved Faraday cup geometry was employed for measurement of fast-rising pulses of energetic beam. In Ref. [13] a versatile Faraday cup was designed and constructed for characterizing of the Z-dependence measurement using Be, C, Al and Pb target plates. In another work, instead of the suppressor grid an absorbing grid of electron was used to derive the secondary electron yield by analyzing the grid and collector signals [14]. In Ref. [15] a nested Faraday probe with two concentric ion collectors and a guard ring was fabricated to investigate the effects of gap configuration between collectors and guard ring on Faraday probe measurements. This paper is organized as follows. Design, optimization and construction of the Faraday cup is presented in Sec. 2. The experimental setup and diagnostics are depicted in Sec. 3. The experimental results are described in Sec. 4, followed by conclusion in Sec. 5.

2. Design, optimization and construction of the Faraday cup

Angular distribution of ion-induced secondary emission follows from the cosine distribution function with its peak in the normal direction of the surface and without dependence on the ion incidence angle [16]. On the other hand, angular distribution of scattered ion emission follows from Gaussian-like distribution with its peak being greatly dependent on the primary ion incidence angle [17–19]. Design

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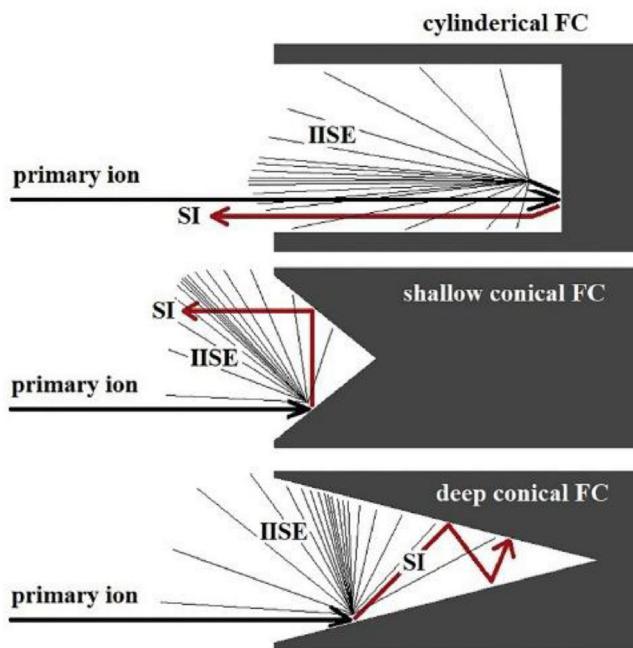


Fig. 1. A simple model for ion-induced secondary electrons (IISEs) and scattered ion (SI) emission due to incidence of a primary ion inside the Faraday cup for three different cavities.

of the Faraday cup was performed in CST STUDIO code [20].

A simple model was presented for comparing three configurations of inner Faraday cup cavity from the point of view of recapture ability of secondary particles inside the cup (Fig. 1). A common cylindrical, a shallow conical, and a deep conical Faraday cup were employed. Based on this qualitative model, in the deep conical configuration the emission angle of the secondary electrons is in the direction of the cup wall. This increases electron absorption probability. It also causes the primary ions to disperse inside the cup and gives more opportunity to neutralize the primary incident ions. Using a negative potential barrier in the entrance of the Faraday cup is an effective way to reduce the loss of the secondary electrons. The scheme of Faraday cup with a suppressor ring in its entrance is shown in Fig. 2. At a constant voltage of -100 V the.

Curve of the electric field along the internal axis of the cup (curve-1) was obtained for the ring lengths in the range of 2–20 mm in steps of 2 mm. Then, the electric field was integrated over curve-1 by Origin software [21] for each length of the ring and presented in Fig. 3. As it is clear, the electric field integration increases rapidly with increasing the ring length from 2 to 10 mm. But, in the ranges of 10–20 mm the curve slope decreases. Therefore, due to the limitation in dimensions of Faraday cup the length of 10 mm was selected for the ring. As shown in Fig. 4, the constructed Faraday cup includes 5 parts. A cylindrical metal body surrounds the ion collector (inner cup) and the suppressor ring to

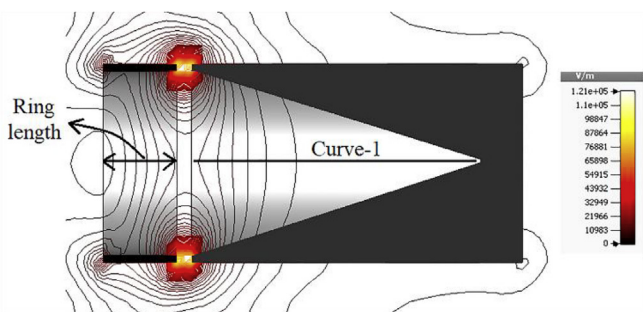


Fig. 2. Scheme of Faraday cup with suppressor ring in entrance.

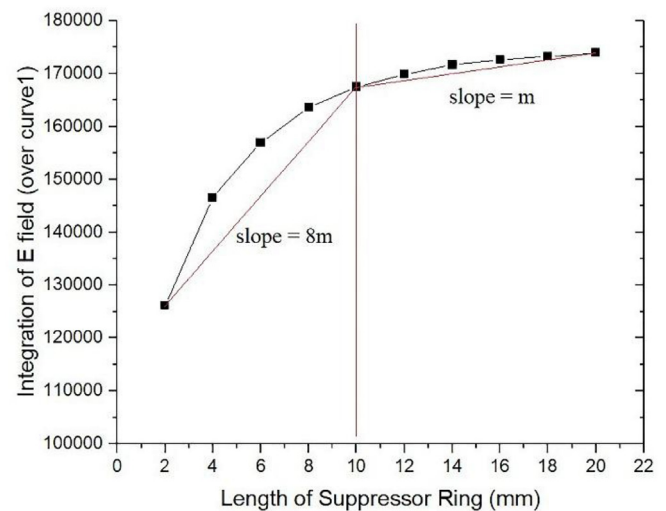


Fig. 3. Electric field integration over the curve-1 for different lengths of ring.

protect them from the peripheral noises. The ion collector is made of a conical cavity at an angle of 26° and an entrance diameter of 25 mm to collect the whole ions. These are insulated from each other by PTFE insulators. The main dimensions and materials of each components are presented in Table 1. The electrical circuit scheme is shown in Fig. 5. The main part of the electronic unit is an AVR micro controller model ATmega16. Its function is to receive analogue data (up to $+5$ V) and convert it to digital data and then transfer it to the LCD and computer. The electronic unit consists of four individual circuits: internal power supply circuit, ion beam current circuit, suppressor ring circuit and thermocouple circuit. The internal power supply circuit is made of a diode bridge, two capacitors ($2200 \mu\text{F}$, 25 V) and three voltage regulators with three different outputs ($+5$ V and ± 12 V). The input of internal power supply is provided by a ± 15 V transformer which is connected to 220VAC and its outputs are $+5$ V for the micro controller and LCD and ± 12 V for op amps. The most important function of the electronic unit is to measure the electrical current of positive ions that are accumulated by the ion collector. The ion beam current circuit consists of three 100Ω , $1 \text{ k}\Omega$ and $10 \text{ k}\Omega$ shunt resistors in parallel which are selected by a multiway switch to measure the wide range of currents. In this work, the $10 \text{ k}\Omega$ resistor was used to measure the maximum current of $500 \mu\text{A}$ with a precision of $1 \mu\text{A}$. In order to reduce the fluctuations of the ions current, a 100 nH inductor was used in the current line before shunt resistor. Also, two 100 nF and $100 \mu\text{F}$ capacitors were placed in parallel with the shunt resistor as the electromagnetic noises filter. The accumulated positive ions in the cup lead to produce an electrical potential (voltage) difference in the shunt resistor which is read by the micro controller and then is transferred to the LCD and computer. The electronic unit was calibrated by applying a specific electrical current to the ion collector. During the measurements a potential difference in shunt resistor was also measured by an oscilloscope (model Tektronix TDS2024C). We observed a deviation of about $\pm 1.5\%$. For measuring the negative voltage of the suppressor ring, first, the voltage must be reduced by a voltage divider and then converted to a positive voltage by a voltage converter op amp.

The maximum voltage that is applied to the ring is -300 V. Therefore, the use of 100Ω and $6 \text{ k}\Omega$ resistors for the voltage divider converts the voltage of -300 V to -5 V. Then, the negative voltage is converted to positive by the LM741 chip. By this way, the voltages of 0 to -300 V are converted to 0 to $+5$ V in proportion to the suppressor ring voltage. This analogue voltage (0 to $+5$ V) is received by a micro controller and after converting to the digital data between 0 and -300 is transmitted for the LCD and computer. A K-type thermocouple is used to control the collector temperature and prevent it from overheating



Fig. 4. Image and drawing of the constructed Faraday cup.

Table 1
Dimensions and materials of each Faraday cup components.

Component	d _{out} (mm)	d _{in} (mm)	L (mm)	material
Body	38	36.5	72	Stainless Steel 304
Cup (ion collector)	30	25 (entry)	50 outside	Copper
Suppressor ring	27	25	10	Stainless Steel 304
Front insulator	36.5	27	14	PTFE
Back insulator	36.5	–	25	PTFE

which can damage the insulators. The thermocouple was mounted on the outer surface of the ion collector and electrically insulated from the collector with a thin layer of silicon paste. When the tip of the thermocouple experiences a change in temperature a voltage about a few hundred μV is created in two wire legs of the thermocouple. It is amplified by the LM741 chip to convert to the detectable voltage for the micro controller. The thermocouple circuit was calibrated by an

Alcohol thermometer. The electronic unit of the Faraday cup is shown in Fig. 6.

3. Experimental setup and diagnostics

The helicon ion source device consists of two main parts. The first part is helicon plasma source and the second part is ion extraction system. The device includes a Pyrex tube that is surrounded by a helix antenna and two electromagnetic coils in Helmholtz configuration. The antenna is connected to a 2 kW, 13.56 MHz RF power supply through an L-type impedance matching network. The electromagnetic coils can be connected to two different current sources which leads to production of magnetic field strength of 380 and 760 G. Diode extraction system was employed for the ion beam generation. The plasma and ground electrodes have a conical geometry with 2 and 5 mm aperture diameters, respectively. The gap distance between the electrodes is 6.5 mm. The extraction chamber is made of stainless steel. The plasma and the

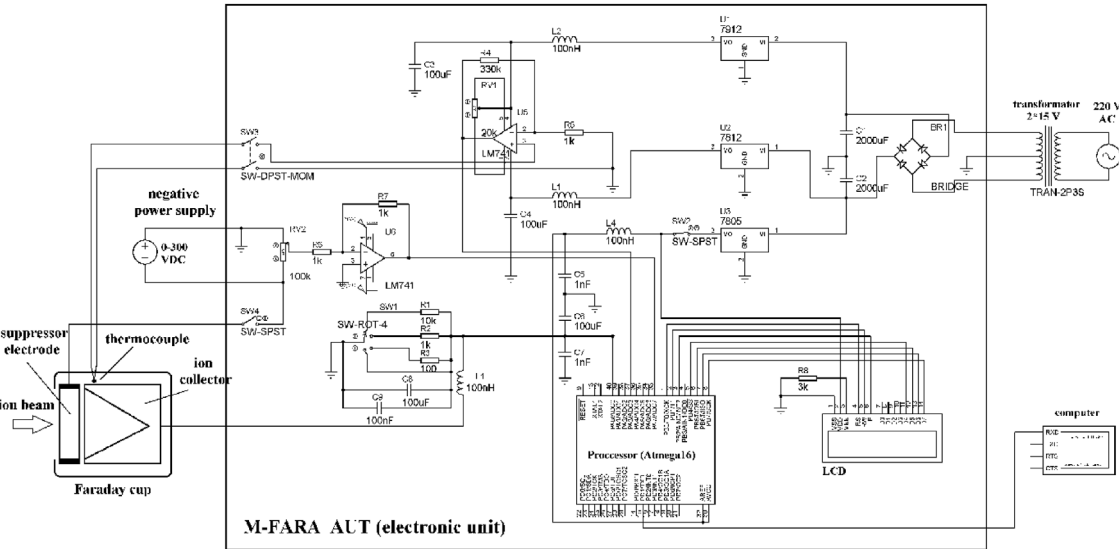


Fig. 5. Electrical circuit scheme of Faraday cup.

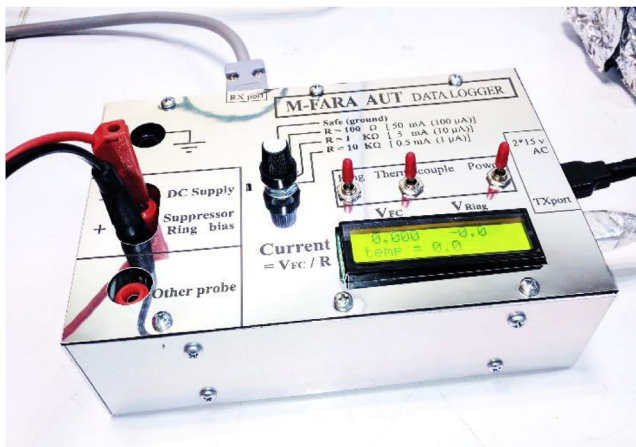


Fig. 6. Electronic unit of Faraday cup.

extractor chamber were evacuated to a base pressure of 10^{-6} Torr using a turbo pump and a rotary pump. The operating pressure was in the range of 10^{-4} – 10^{-3} Torr. High voltage is applied to the plasma electrode by DC power supply, which has a maximum voltage up to 40 kV. More description of the constructed helicon ion source can be found in Ref. [1]. A scheme of the device is shown in Fig. 7. In this experiment, RF power and the extraction voltage increase up to 1300 W and 20 kV, respectively. The Faraday cup was placed at a distance of 10 cm from the ion extraction system to measure the total ions current. In addition, optical emission spectroscopy (OES) method employed for demonstration of the mode transition from inductively coupling plasma (ICP) to helicon. A pair of lenses were used for gathering the radiation from the center of plasma. The received radiation transmitted to spectrometer and then recorded in the computer.

4. Result and discussion

To determine the needed bias between the ring and the Faraday cup collector a test has made where the Faraday cup output was measured

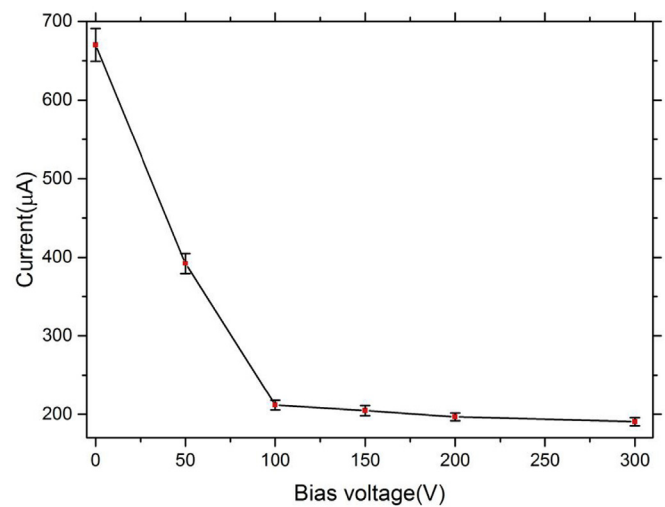


Fig. 8. The decrease in registered ion beam on the Faraday cup as its bias increases at 15 kV extraction voltage, 760 G magnetic field strength, 0.83 mTorr working pressure and 1000 W RF power.

for different biases up to 300 V, keeping the rest of the setup constant. As seen from Fig. 8, the measured ion beam current saturates for biasing voltages closed to 100 V. Therefore, a negative voltage of 100 V will apply to the suppressor ring to prevent the loss of secondary electrons in the rest of the context. This is in agreement with Ref. [16], where the maximum energy of the secondary electrons that is emitted from Faraday cup, is less 100 eV for a 40 kV Ar^{1+} ion beam [16]. In this paper, for each operating condition five CW discharges were performed. The error bars are defined as the value of the measured ion beam which deviates from mean values. The reproducibility of the current measurements under same conditions is within 2–4%. This error is due to accumulation of the experimental setup errors like instabilities, RF noise, variation in plasma parameters, working pressure and gas flow rate. An important parameter for the ion current is the extraction voltage. According to the Childe-Langmuir equation [22,23], in a simple

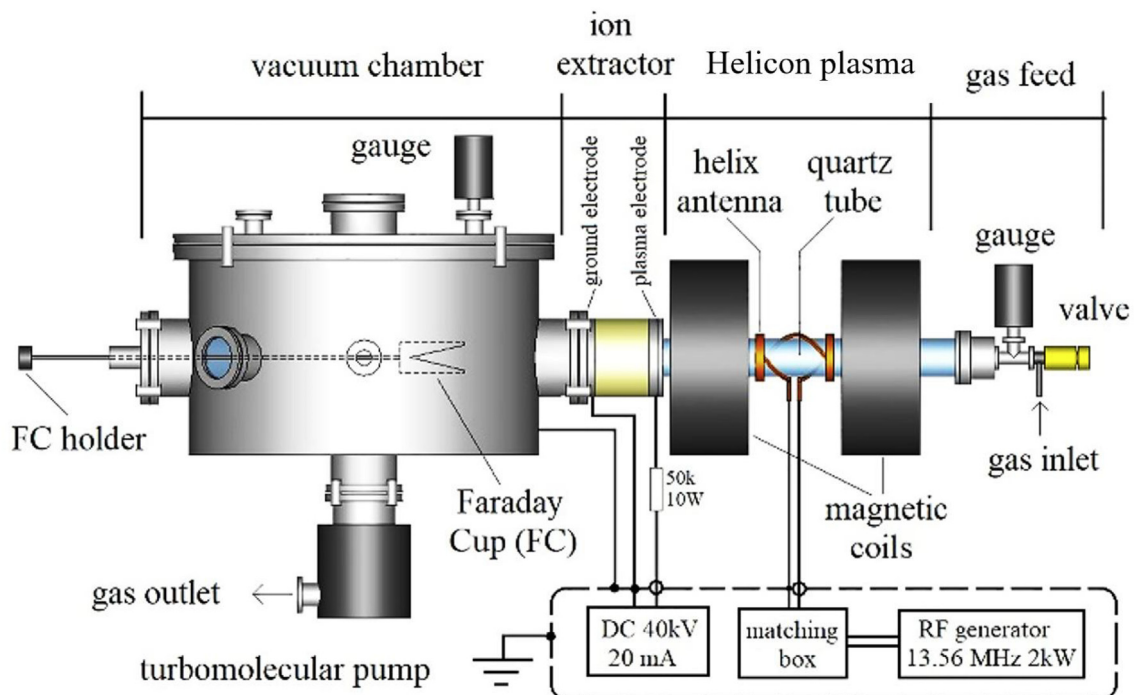


Fig. 7. Scheme of the constructed helicon ion source device [1].

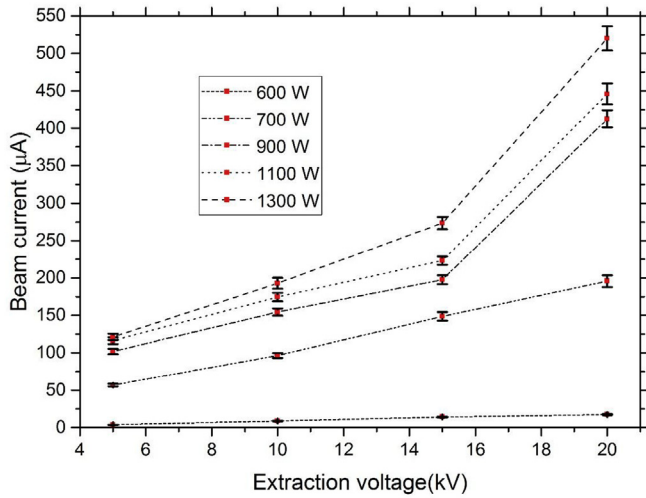


Fig. 9. Effect of RF power and extraction voltage on the measured ion beam current. Operating pressure is 0.83 mTorr.

two-electrode extractor, the beam current is proportional to $V^{3/2}$, where V is the extraction voltage:

$$J = \frac{4\epsilon_0}{9} \sqrt{\frac{2q}{M}} \frac{V^{3/2}}{d^2} \quad (1)$$

Where ϵ_0 is permittivity tensor, q is the ion charge, V is the extraction voltage, M is the ion mass and d is the electrode gap. Child-Langmuir law is satisfied in certain conditions, as in.

[24] the Child-Langmuir law is valid at a voltage between 6 and 15 kV. In another work, it was observed that the Child-Langmuir curve is followed up to 1 kV extraction voltage [25]. As shown in Fig. 9, the ion beam current was measured using the Faraday cup as the extraction voltage was varied between 5 and 20 kV at different RF powers. The measured ion beam current increases with enhancing the RF power and extraction voltage. The measured curves seem to scale well with $V^{3/2}$. Because of the source design we haven't become near to the space charge limit yet. In Ref. [25] they reported that with an increase in the extraction potential, the plasma meniscus starts to become concave which increases the area of emission and the meniscus moves away from the anode and thereby the I–V characteristics deviates significantly from Child-Langmuir curve. The constructed ion source may work in CCP, ICP and helicon mode at different operation conditions like the amount of the RF power and magnetic field strength. One of the methods for showing the mode transition from ICP to helicon is optical emission spectroscopy (OES) [1,26]. In this work, when the magnets was turned off the argon plasma was operating in the ICP mode and when the plasma column was magnetized the helicon mode discharge reached for RF powers more than 700 W. A typical plasma spectrum in the ICP and helicon modes at 1000 W RF power is depicted in Fig. 10. The application of the magnetic field leads to an increase in the intensity of spectrums, which indicate the increase of the plasma density. Also, some spectrums (such as 434.8, 442.6, 460.95, 473.59, 480.6 and 487.98 nm Ar II line emissions) are clearly observed only in the helicon mode. The value of ion beam current at 15 kV extraction voltage at different RF powers is illustrated in Fig. 11. A jump in the ion beam current was observed at 700 W RF power. This also means that the mode transition from ICP to helicon reaches at this power.

5. Conclusion

An optimized Faraday cup was made with an interior conical geometry to reduce the loss of secondary electrons and backscattered ions. The collector was made of copper while the suppressor ring and the Faraday cup body were made of stainless steel. In addition, a biased

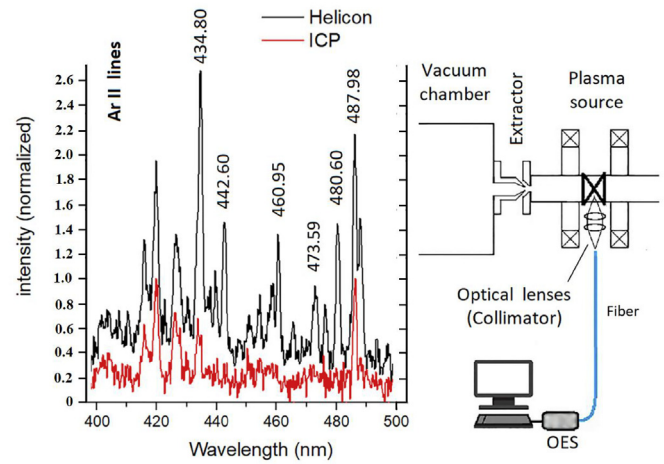


Fig. 10. Optical spectroscopy of plasma in the ICP and helicon modes at 1000 W RF power, 760 G magnetic field and 1.4 mtorr working pressure (The results are normalized to 420 nm line at ICP mode).

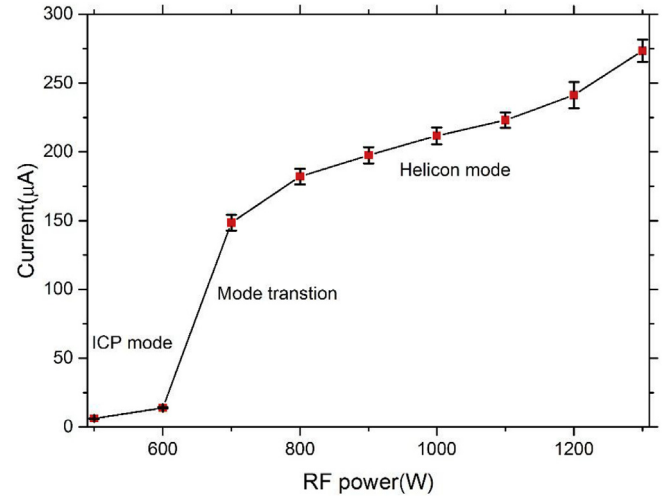


Fig. 11. Demonstration of the mode transition from ICP to helicon through ion beam measurement at 15 kV extraction voltage and 0.83 mTorr operating pressure.

suppressor ring was placed at the collector entrance to recapture low energy electrons. Faraday cup output was measured for different biases up to 300 V, keeping the rest of the setup constant. The measured ion beam current saturates for biasing voltages closed to 100 V. An electronic unit was made for measuring the continuous ion current, suppressor ring voltage, and temperature of ion collector. CST studio code was used for design of the Faraday cup. A common cylindrical, a shallow conical, and a deep conical Faraday cup were compared. Based on this qualitative model, in the deep conical configuration the emission angle of the secondary electrons is in the direction of the cup wall, which increases their absorption probability. Also, it causes the primary ions to disperse inside the cup, which gives more opportunity to neutralize the primary incident ions. The electric field integration increases rapidly with increasing the ring length from 2 to 10 mm, but in lengths of 10–20 mm, the curve slope decreases. Therefore, due to the limitation in dimensions of Faraday cup, the length of 10 mm was selected for the ring. The measured ion beam current and the Ar– II line emission indicate that the constructed ion source works in the helicon mode at RF powers more than 700 W. The measured ion beam curves seem to scale well with $V^{3/2}$.

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References

- [1] M. Afsharmanesh, M. Habibi, Development of a helicon ion source: simulations and preliminary experiments, *Rev. Sci. Instrum.* 89 (3) (2018) 033301.
- [2] E. Cantero, et al., Design of a compact Faraday cup for low energy, low intensity ion beams, *Nucl. Instrum. Methods Phys. Res. Sect. A Accel. Spectrom. Detect. Assoc. Equip.* 807 (2016) 86–93.
- [3] J.S. Pearlman, Faraday cups for laser plasmas, *Rev. Sci. Instrum.* 48 (8) (1977) 1064–1067.
- [4] J. Prokúpek, et al., Development and first experimental tests of Faraday cup array, *Rev. Sci. Instrum.* 85 (1) (2014) 013302.
- [5] G. Roshani, M. Habibi, M. Sohrabi, An improved design of Faraday cup detector to reduce the escape of secondary electrons in plasma focus device by COMSOL, *Vacuum* 86 (3) (2011) 250–253.
- [6] P. Strehl, *Beam Instrumentation and Diagnostics* vol. 120, Springer, 2006.
- [7] J. Thomas, et al., Performance enhancement study of an electrostatic Faraday cup detector, *Nucl. Instrum. Methods Phys. Res. Sect. A Accel. Spectrom. Detect. Assoc. Equip.* 536 (1–2) (2005) 11–21.
- [8] K. Brown, G. Tautfest, Faraday-cup monitors for high-energy electron beams, *Rev. Sci. Instrum.* 27 (9) (1956) 696–702.
- [9] A. Barnett, S. Olbert, Response function of modulated grid Faraday cup plasma instruments, *Rev. Sci. Instrum.* 57 (10) (1986) 2432–2440.
- [10] A. Case, et al., Designing a sun-pointing Faraday cup for solar probe plus, *AIP Conference Proceedings*, AIP, 2013.
- [11] H. Chuaqui, et al., Simple Faraday cup with subnanosecond response, *Rev. Sci. Instrum.* 60 (1) (1989) 141–142.
- [12] D. Pellinen, A high current, subnanosecond response Faraday cup, *Rev. Sci. Instrum.* 41 (9) (1970) 1347–1348.
- [13] J. Seamans, W. Kimura, Design and characterization of a versatile Faraday cup, *Rev. Sci. Instrum.* 64 (2) (1993) 460–469.
- [14] H. Kelly, et al., Analysis of the nitrogen ion beam generated in a low-energy plasma focus device by a Faraday cup operating in the secondary electron emission mode, *IEEE Trans. Plasma Sci.* 26 (1) (1998) 113–117.
- [15] D.L. Brown, A.D. Gallimore, Evaluation of ion collection area in Faraday probes, *Rev. Sci. Instrum.* 81 (6) (2010) 063504.
- [16] J. Mischler, et al., Angular distributions of secondary electrons emitted in Ar + -polycrystalline Al collisions, *Surf. Sci.* 136 (2–3) (1984) 532–544.
- [17] E. Formann, F. Viehböck, H. Wotke, Anomalous angular distribution of scattered heavy ions, *Phys. Lett.* 23 (9) (1966) 558–559.
- [18] B.J. Garrison, Theory of ion scattering from single crystals, *Surf. Sci.* 87 (2) (1979) 683–699.
- [19] A. Marwick, et al., The low-angle scattering of energetic ions from an amorphous surface, *Radiat. Eff.* 10 (1–2) (1971) 49–54.
- [20] CST, <<http://www.cst.com>>.
- [21] Origin, <https://www.originlab.com>.
- [22] I.G. Brown, *The Physics and Technology of Ion Sources*, John Wiley & Sons, 2004.
- [23] W. Stirling, C. Tsai, P. Ryan, 15 cm duoPIGatron ion source, *Rev. Sci. Instrum.* 48 (5) (1977) 533–536.
- [24] A. Pysznik, et al., Extraction of ions from a plasma source and formation of beams, *Instrum. Exp. Tech.* 50 (4) (2007) 552–556.
- [25] P. Nabhiraj, et al., Low energy micron size beam from inductively coupled plasma ion source, *J. Vac. Sci. Technol. B, Nanotechnol. Microelectron.: Mater. Process. Meas. Phenom.* 29 (5) (2011) 051604.
- [26] M. Celik, Spectral measurements of inductively coupled and helicon discharge modes of a laboratory argon plasma source, *Spectrochim. Acta, Part B* 66 (2011) 149–155.